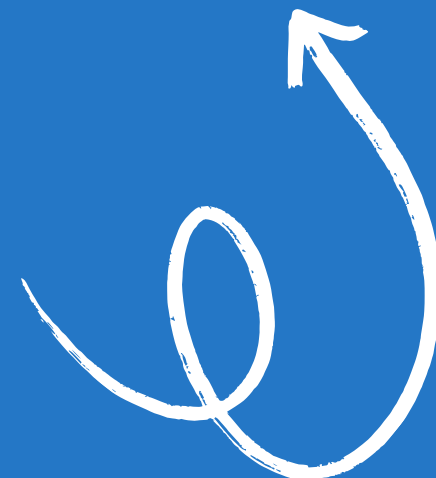


Baba Banda Singh Engineering College

Fuel Efficiency Prediction

Project team:

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Sahil Dhiman



Introduction

- ML-based system to predict vehicle fuel efficiency
- Trains multiple regression models for accuracy
- Streamlit web app for real-time user input and prediction
- Results displayed in MPG and km/l
- Supports fuel-efficient and eco-friendly decisions

Objectives

- To develop a machine learning model that accurately predicts vehicle fuel efficiency.
- To analyze the relationship between vehicle features and fuel consumption.
- To display results in both MPG and km/l for better usability in India.
- To support environment-friendly and fuel-efficient driving decisions.

Literature Survey

- ML outperforms traditional statistical methods
- Handles complex, non-linear relationships in data
- Regression models widely used (Linear, Random Forest, Gradient Boosting)
- Auto MPG is a common benchmark dataset
- Feature engineering improves model accuracy

Methodology



Data Collection
& Acquisition



Data Cleaning
&
Preprocessing



Feature
Engineering &
Selection



Model Training
& Optimization



Model Evaluation &
Comparison

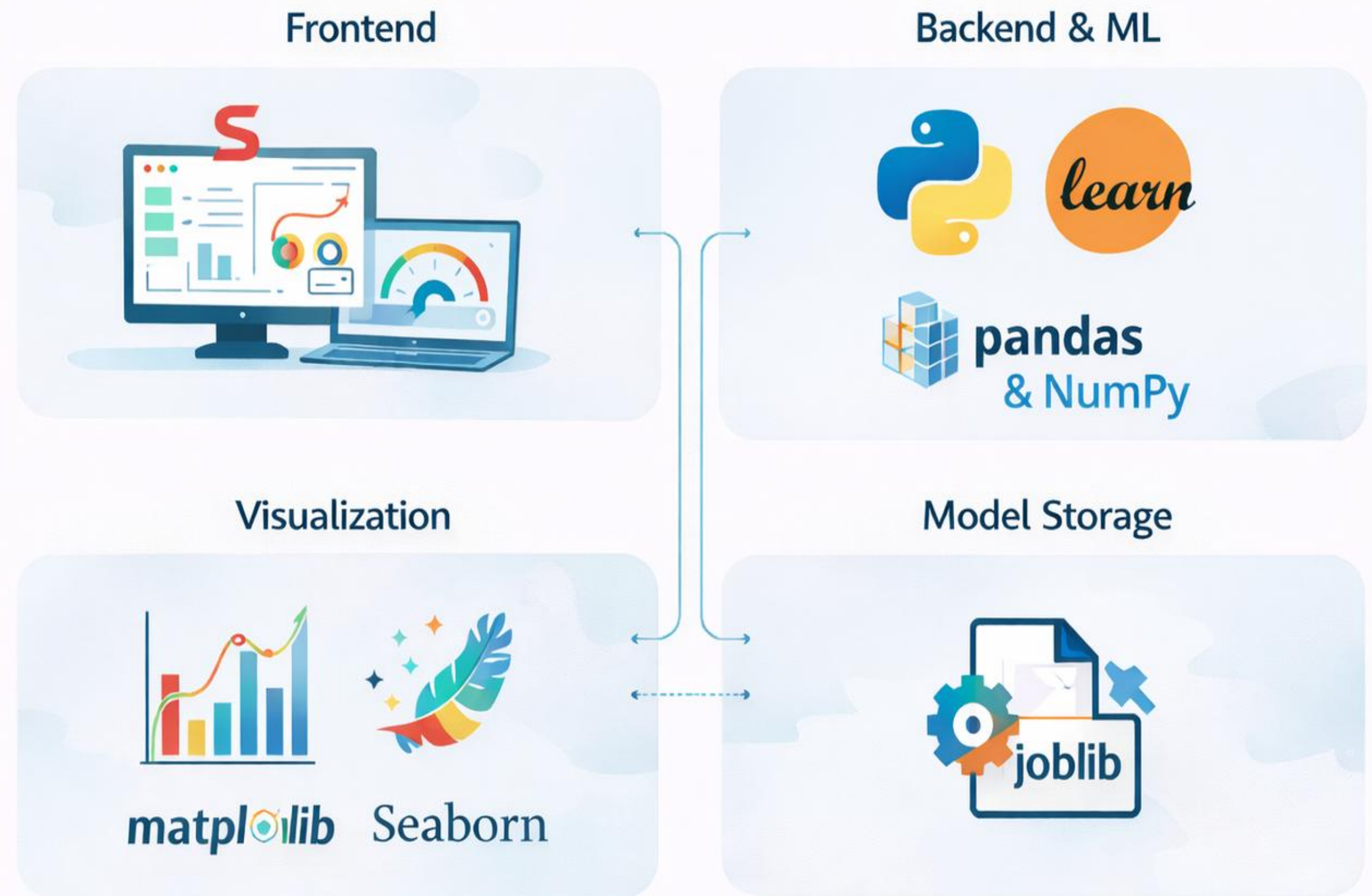


Model Saving &
Versioning



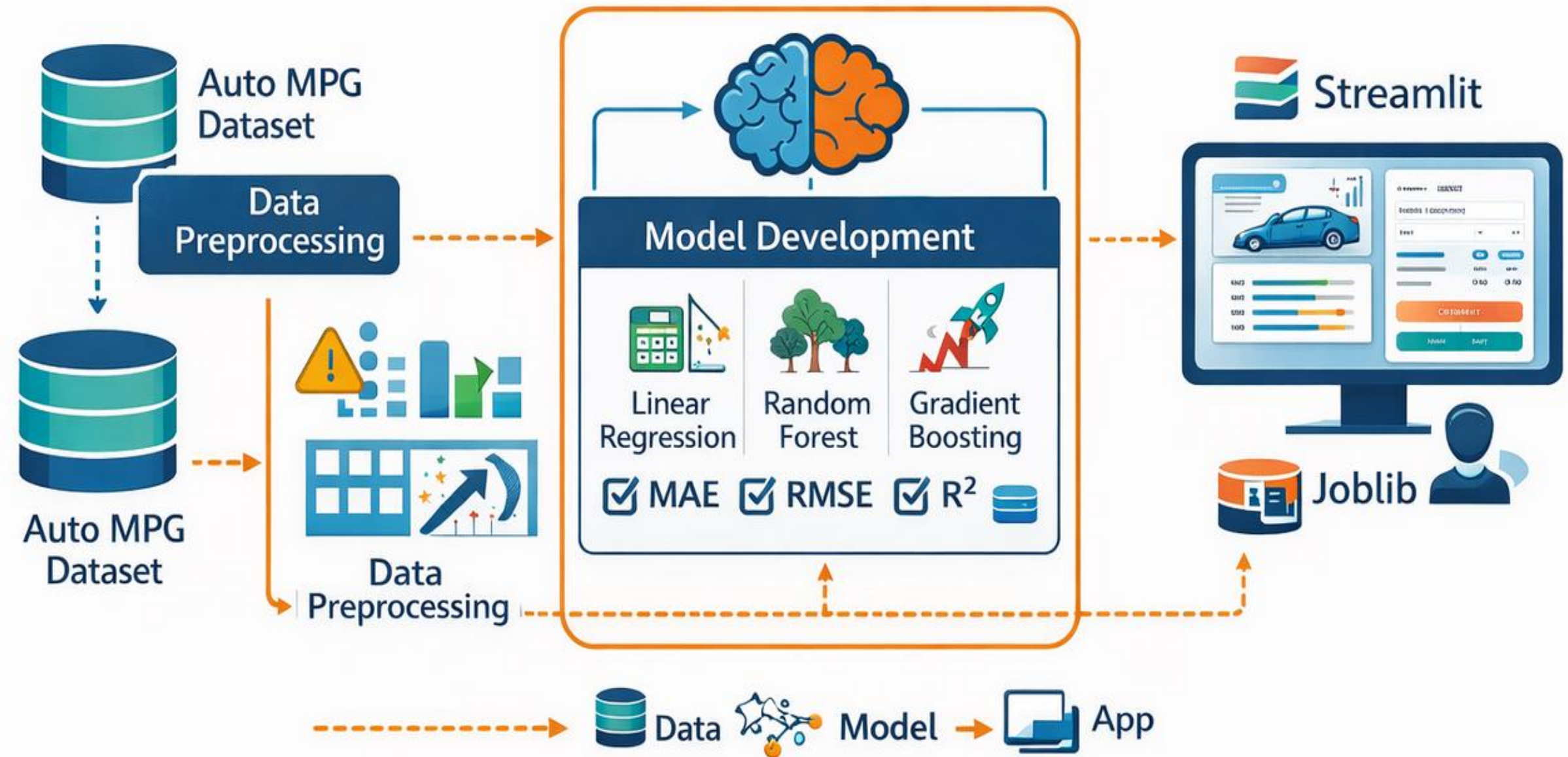
Streamlit Application
Deployment

Technologies Used



System Architecture Overview

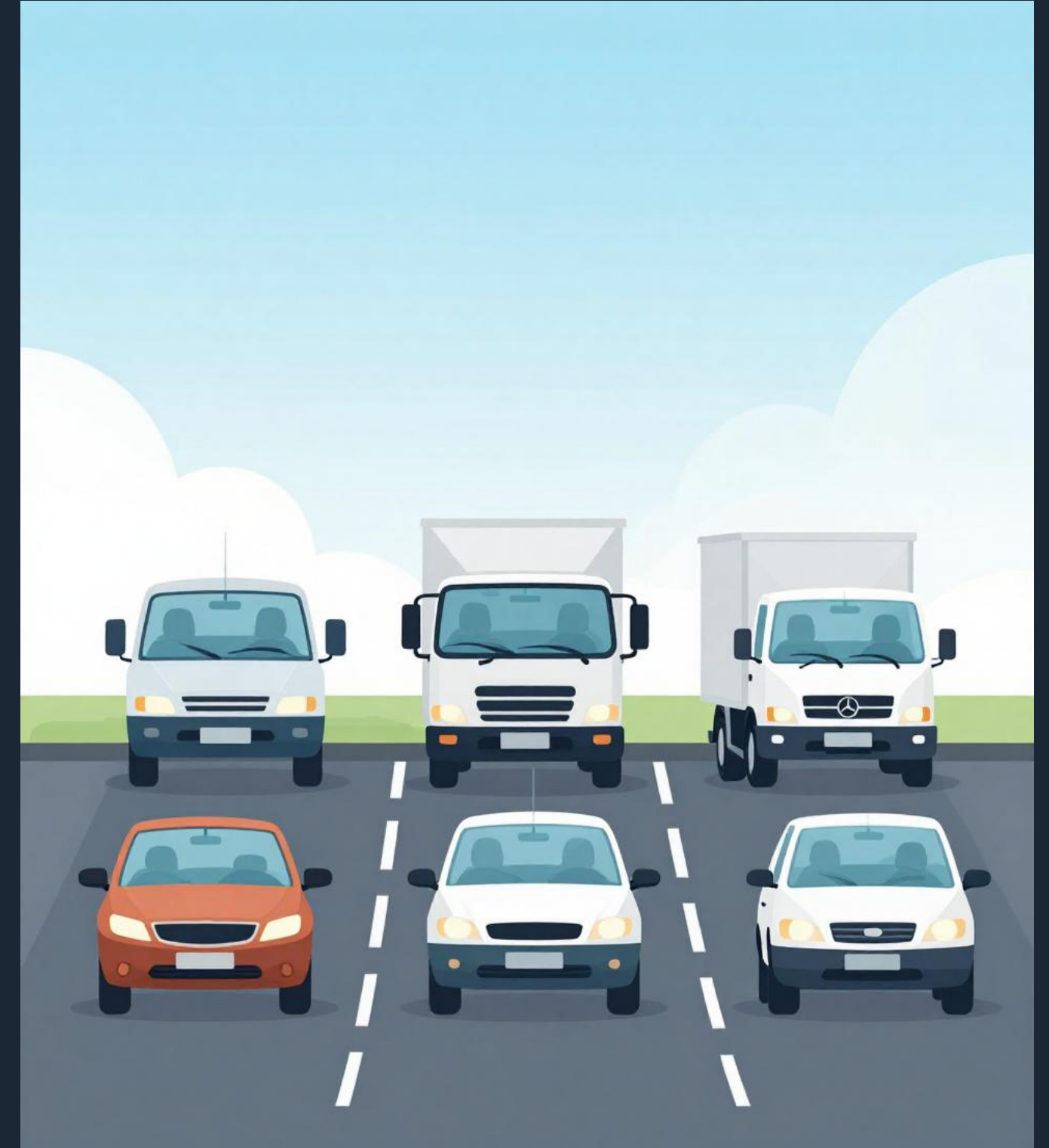
System Architecture Overview



- Uses a layered architecture: Data → Model → App
- Data preprocessing: missing values, encoding, scaling
- Feature engineering: correlation & feature selection
- Evaluate using MAE, RMSE, R² and save best model (Joblib)

Outcomes & Future Impact of Fuel Efficiency

- Accurate prediction of vehicle fuel efficiency
- User-friendly Streamlit web application for real-time prediction
- Visual graphs for better interpretation of results
- Reliable model saved using Joblib for reuse
- Lower carbon emissions, helping the environment
- Reduced fuel consumption and cost for users



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Thank You